

TRAVAIL PRATIQUE #1

OBJECTIFS

- Partitionnement et système de fichier Linux.
- Gestion de stockage
- Adressage IP et segmentation en sous-réseau.
- Configurer les paramètres réseau d'une station de travail Linux.

CONSIGNES

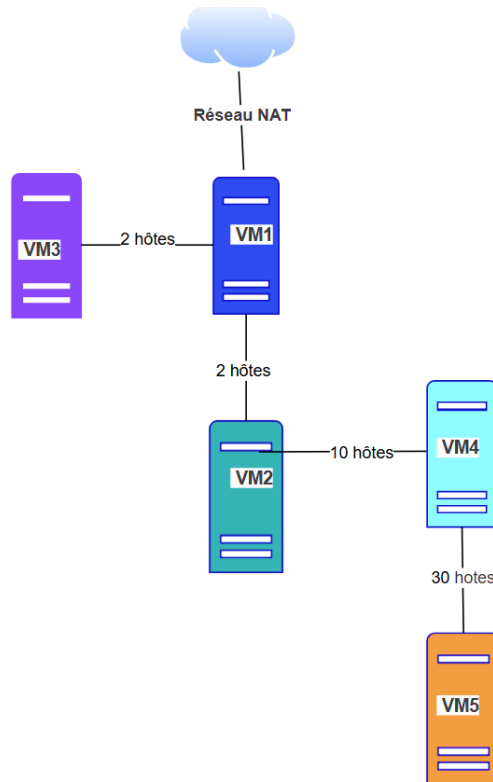
- Le TP 1 est sommatif noté sur 100 et vaut **20% de la note finale**.
- Vous devez remettre un **rapport détaillé** des commandes appliquées pour chaque partie, des résultats obtenus et des remarques s'il y a lieu. (Utiliser des **captures d'écran bien visibles**).
- Le rapport est à terminer et à remettre avant la date limite. La remise de votre rapport de laboratoire doit se faire via Moodle. Le fichier rapport doit être nommé **Prénom_Nom_Rapport_TP1_3G5.pdf**.
- Le travail pratique se fait individuellement. Chacun doit démontrer et expliquer l'application des paramètres et configurations exigées. Des questions vous seront posées **lors de la validation en classe**.
- **Le rapport est à remettre au plus tard le 2 novembre 2025 à 23h59. La validation du travail se fera en classe le lundi 3 novembre 2025, selon un ordre préétabli par votre enseignante.**

PONDÉRATION

Procédure/partie	Noté sur
Partie 1 - Installation - partitionnement	15%
Partie 1 - validation	5%
Partie 2 - Schéma d'adressage	15%
Partie 2 - validation	5%
Partie 3 - Routage	20%
Partie 3 - validation	10%
Partie 4 - défi	10%
Partie 4 - validation	5%
Qualité du rapport (respect des exigences, ...)	10%
Qualité du français	5%
Total / 100	100%

TRAVAIL À RÉALISER

PARTIE 1 – INSTALLATION – PARTITIONNEMENT



1- TOPOLOGIE :

2- EXIGENCES :

Créez les machines virtuelles de la topologie en respectant les spécifications suivantes :

Machine	Système d'exploitation	Nom de la VM	Nom d'hôte	Mémoire	Taille disque	Type installation
VM1	AlmaLinux	VM1	VM1	1 Go	10 Go	Minimale
VM2	Clone de la VM1	VM2	VM2	1 Go	10 Go	Minimale
VM3	Clone de la VM1	VM3	VM3	1 Go	10 Go	Minimale
VM4	Clone de la VM1	VM4	VM4	1 Go	10 Go	Minimale

VM5	Clone de la VM1	VM5	VM5	1 Go	10 Go	Minimale
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CONFIGURATION DE LA VM1 :

Device	Summary
Memory	1 GB
Processors	2
Hard Disk (SATA)	10 GB
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

Notes importantes :

- AlmaLinux OS 9.6 Minimal ISO

https://repo.almalinux.org/almalinux/9.6/isos/x86_64/AlmaLinux-9.6-x86_64-minimal.iso

- Lors de l'installation et de la configuration de votre VM AlmaLinux, vous devez créer un utilisateur avec votre Nom et prenom

- Effectuez toutes les étapes de ce TP en utilisant le compte **root**.

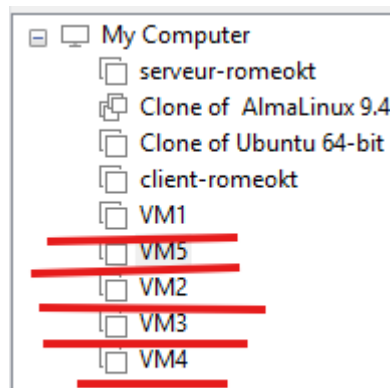
```
[root@VM1 ~]#
```

- Après avoir installé la **VM1**, et **avant de cloner les autres VMs**, exécutez cette commande pour installer les paquets suivants : **dnf -y install vim lvm2 bash-completion**

```
[root@VM1 ~]# dnf -y install vim lvm2 bash-completion_
```

```
Installed:
bash-completion-1:2.11-5.el9.noarch
device-mapper-persistent-data-1.1.0-1.el9.x86_64
libnvm2-1.11.1-1.el9.x86_64
lvm2-libs-9:2.03.28-6.el9.x86_64
pkgconf-pkg-config-1.7.3-10.el9.x86_64
vim-filesystem-2:8.2.2637-22.el9_6.1.noarch
device-mapper-event-9:1.02.202-6.el9.x86_64
gpm-libs-1.20.7-29.el9.x86_64
libpkgconf-1.7.3-10.el9.x86_64
pkgconf-1.7.3-10.el9.x86_64
vim-common-2:8.2.2637-22.el9_6.1.x86_64
device-mapper-event-libs-9:1.02.202-6.el9.x86_64
libaio-0.3.111-13.el9.x86_64
lvm2-9:2.03.28-6.el9.x86_64
pkgconf-m4-1.7.3-10.el9.noarch
vim-enhanced-2:8.2.2637-22.el9_6.1.x86_64

Complete!
```



CLONAGE DES VMS :

CHANGER LE NOMS DES HOTES DES CLONES (JAI FAIT CELA POUR CHAQUE CLONES VM1 À VM5) :

```
[root@VM1 ~]# sudo hostnamectl set-hostname VM5
```

3- PARTITIONNEMENT :

PARTITIONNEMENT À APPLIQUER LORS DE L'INSTALLATION DES SYSTÈMES D'EXPLOITATION VM1:

Point de montage	Taille	Système de fichiers	Type de partition
swap	1 Go	swap	Standard
/boot	1 Go	xfs	Standard
/	5 Go	ext4	Standard
/var	3 Go	ext4	Standard

Vérification des partitions créés durant la configuration :

```
[root@VM1 ~]# df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0    4.0M   0% /dev
tmpfs           tmpfs     870M   0    870M   0% /dev/shm
tmpfs           tmpfs     348M   5.1M 343M   2% /run
/dev/sda2       ext4      4.9G   1.2G 3.5G   26% /
/dev/sda3       ext4      2.9G   144M 2.6G    6% /var
/dev/sda1       xfs       960M   233M 728M   25% /boot
tmpfs           tmpfs     174M   0    174M   0% /run/user/0
```

```
[root@VM1 ~]# swapon --show
NAME      TYPE      SIZE USED PRIO
/dev/sda5 partition 1023M  0B   -2
```

PARTITIONNEMENT À APPLIQUER APRÈS L'INSTALLATION DES SYSTÈMES D'EXPLOITATION

VM3: Rattachez un nouveau disque **SATA** de **3.5 Go** à la **VM3**.

Device	Summary
Memory	2 GB
Processors	2
Hard Disk (SATA)	10 GB
New Hard Disk (SATA)	3.5 GB
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

- Créer une partition 1 de **2 Go** en **ext2**, montée dans le dossier **/Part1** (montage permanent).
Créer la partition :

```
# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x70a785b6.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-7340031, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-7340031, default 7340031): +2

Created a new partition 1 of type 'Linux' and of size 2 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
[ 74.056730] sdb: sdb1
Syncing disks.
```

Formater la partition en ext2:

```
# mkfs.ext2 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 2c88923f-6c73-4602-969d-e4ce4918b441
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Writing superblocks and filesystem accounting information: done
```

Création du point de montage:

```
# mkdir /Part1
```

Rendre le montage permanent :

```
#
# /etc/fstab
# Created by anaconda on Tue Oct 28 16:28:06 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8979f865-68fe-498d-986d-b6eab3450e84 / ext4 defaults 1 1
UUID=1d216cbb-b3cb-4573-bc27-c3f4463a7edc /boot xfs defaults 0 0
UUID=5a7edcd7-2426-439e-9d23-836816b7b0a5 /var ext4 defaults 1 2
UUID=1c3d1e00-13b9-4c82-ba28-c79da2b40db6 none swap defaults 0 0
UUID=2c88923f-6c73-4602-969d-e4ce4918b441 /Part1 ext2 defaults 0 0
```

Faire le montage :

```
# mount -a
[ 1320.677662] EXT4-fs (sdb1): mounting ext2 file system using the ext4 subsystem
[ 1320.680964] EXT4-fs (sdb1): mounted filesystem 2c88923f-6c73-4602-969d-e4ce4918b441 r/w without journal. Quota mode: none.
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
```

Vérification :

```
# df -h /Part1
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.0G   24K  1.9G   1% /Part1
```

- Créer une partition 2 de **1 Go** en **ext4**, monté dans le dossier **/Part2** (montage permanent).

Créer la partition :

```
# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to unmount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): n
Partition type
  p   primary (1 primary, 0 extended, 3 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (4196352-7340031, default 4196352):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (4196352-7340031, default 7340031): +1G
Last sector, +/-sectors or +/-size(K,M,G,T,P) (4196352-7340031, default 7340031): +1G

Created a new partition 2 of type 'Linux' and of size 1 GiB.

Command (m for help): w
The partition table has been altered.
Syncing disks.
```

Formater la partition en ext4:

```
# mkfs.ext4 /dev/sdb2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: b0ad20c1-637c-4335-831e-bc97eb877ca0
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

Création du point de montage:

```
# mkdir /Part2
```

Rendre le montage permanent :

```
#
# /etc/fstab
# Created by anaconda on Tue Oct 28 16:20:06 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8979f865-68fe-498d-986d-b6eab3450e84 / ext4 defaults 1 1
UUID=1d216cbb-b3cb-4573-bc27-c3f4463a7edc /boot xfs defaults 0 0
UUID=5a7edcd7-2426-439e-9d23-836816b7b0a5 /var ext4 defaults 1 2
UUID=1c3d1e00-13b9-4c82-ba28-c79da2b40db6 none swap defaults 0 0
UUID=2c88923f-6c73-4602-969d-e4ce4918b441 /Part1 ext2 defaults 0 0
UUID=b0ad20c1-637c-4335-831e-bc97eb877ca0 /Part2 ext4 defaults 0 0
```

Faire le montage :

```
# mount -a
[ 2660.621303] EXT4-fs (sdb2): mounted filesystem b0ad20c1-637c-4335-831e-bc97eb877ca0 r/w with ordered data mode. Quota mode: none.
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
```

Vérification :

```
# df -h /Part2
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sdb2	974M	24K	907M	1%	/Part2

- Donnez le nombre d'inodes et le nombre de blocs de chaque partition.


```
# df -hi /Part1
Filesystem      Inodes  IUsed  IFree  IUse% Mounted on
/dev/sdb1       128K    11    128K    1% /Part1

# df -hi /Part2
Filesystem      Inodes  IUsed  IFree  IUse% Mounted on
/dev/sdb2       64K     11    64K    1% /Part2

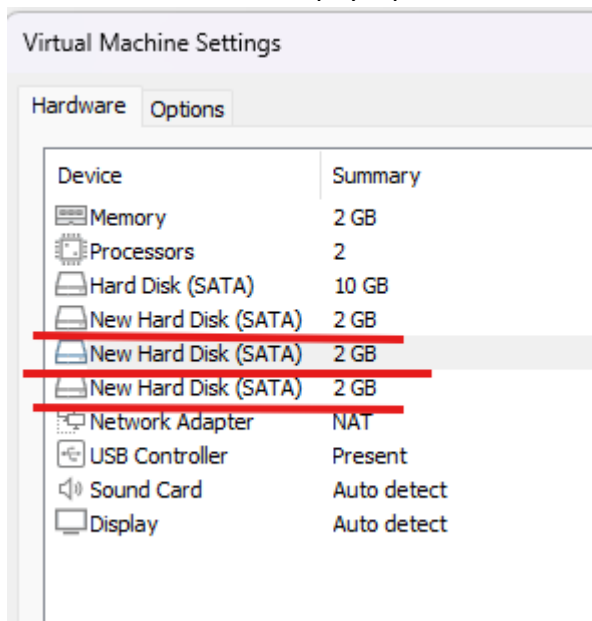
# df -hT /Part1
Filesystem      Type     Size  Used Avail Use% Mounted on
/dev/sdb1       ext2    2.0G   24K  1.9G   1% /Part1

# df -hT /Part2
Filesystem      Type     Size  Used Avail Use% Mounted on
/dev/sdb2       ext4    974M   24K  907M   1% /Part2

# _
```

VM4: Rattachez trois nouveaux disques **SATA** de **2 Go** chaque à la **VM4**.

- Créer trois volumes physiques.



```
# pvcreate /dev/sdb
Physical volume "/dev/sdb" successfully created.
Creating devices file /etc/lvm/devices/system.devices

# pvcreate /dev/sdc
Physical volume "/dev/sdc" successfully created.

# pvcreate /dev/sdd
Physical volume "/dev/sdd" successfully created.
```

Vérification :

```
[root@VM4 ~]# pvs
PV          VG Fmt Attr PSize PFree
/dev/sdb    lvm2 --- 2.00g 2.00g
/dev/sdc    lvm2 --- 2.00g 2.00g
/dev/sdd    lvm2 --- 2.00g 2.00g
```

- Créer un groupe de volume nommé **VM4_VG** avec ces trois nouveaux volumes physiques.

```
# vgcreate VM4_VG /dev/sdb /dev/sdc /dev/sdd
```

```
# vgs
VG      #PV #LV #SN Attr   VSize  VFree
UM4_UG  3    0    0 wz--n- <5.99g <5.99g
```

- Créer LV1 de 2 Go en xfs, montée dans le dossier /Part1 (montage permanent).
- Créer LV2 de 1 Go en ext4, monté dans le dossier /Part2 (montage permanent).

Créer LV1 et LV2:

```
# lvcreate -L 2G -n LV1 UM4_UG
Logical volume "LV1" created.
# lvcreate -L 2G -n LV2 UM4_UG
Logical volume "LV2" created.
```

Vérification :

```
# lvs
LV      VG      Attr      LSize Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
LV1     UM4_UG  -wi-a----- 2.00g
LV2     UM4_UG  -wi-a----- 2.00g
```

Formater LV1 en xfs et LV2 en ext4:

```
# mkfs.xfs /dev/UM4_UG/LV1
meta-data=/dev/UM4_UG/LV1      isize=512    agcount=4, agsize=131072 blks
        =                       sectsz=512    attr=2, projid32bit=1
        =                       crc=1        finobt=1, sparse=1, rmapbt=0
        =                       reflink=1     bigtime=1 inobtcount=1 nrext64=0
data      =                       bsize=4096   blocks=524288, imaxpct=25
        =                       sunit=0      swidth=0 blks
naming    =version 2             bsize=4096   ascii-ci=0, ftype=1
log        =internal log         bsize=4096   blocks=16384, version=2
        =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime   =none                 extsz=4096   blocks=0, rtextents=0

# mkfs.ext4 /dev/UM4_UG/LV2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 88842fc5-eaf7-4c9c-9154-feeb3e3eb242
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Création des points de montage:

```
[root@UM4 ~]# mkdir /Part1
```

```
[root@UM4 ~]# mkdir /Part2
```

Faire le montage de LV1 et LV2:

```
# mount /dev/UM4_UG/LV1 /Part1
[ 1874.793490] XFS (dm-0): Mounting U5 Filesystem 25e2ec03-5542-4f11-8f43-06faf1f107a8
[ 1874.807187] XFS (dm-0): Ending clean mount
# mount /dev/UM4_UG/LV2 /Part2
[ 1886.555017] EXT4-fs (dm-1): mounted filesystem 88842fc5-eaf7-4c9c-9154-feeb3e3eb242 r/w with ordered data mode. Quota mode: none.
```

Rendre le montage permanent :

```
#
# /etc/fstab
# Created by anaconda on Tue Oct 28 16:20:06 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8979f865-68fe-498d-986d-b6eab3450e84 / ext4 defaults 1 1
UUID=1d216cbb-b3cb-4573-bc27-c3f4463a7edc /boot xfs defaults 0 0
UUID=5a7edcd7-2426-439e-9d23-836816b7b0a5 /var ext4 defaults 1 2
UUID=1c3d1e00-13b9-4c82-ba28-c79da2b40db6 none swap defaults 0 0
UUID=88842fc5-eaf7-4c9c-9154-feeb3e3eb242 /Part2 ext4 defaults 0 2
~
~
~
~
~
~
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~
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~
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~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

Faire le montage :

```
[root@VM4 ~]# mount -a
```

- Étendre **LV2** d'un autre **2 Go**.

Démonter /Part2 pour pouvoir faire la modification:

```
[root@VM4 ~]# umount /Part2
```

Étendre LV2:

```
# lvresize --resizefs --size 3G /dev/UM4_UG/LV2
File system ext4 found on UM4_UG/LV2.
File system fsck will be run before extend.
Size of logical volume UM4_UG/LV2 changed from 2.00 GiB (512 extents) to 3.00 GiB (768 extents).
Extending file system ext4 to 3.00 GiB (3221225472 bytes) on UM4_UG/LV2...
e2fsck /dev/UM4_UG/LV2
/dev/UM4_UG/LV2: 11/131072 files (0.0% non-contiguous), 26156/524288 blocks
e2fsck done
resize2fs /dev/UM4_UG/LV2
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/UM4_UG/LV2 to 786432 (4k) blocks.
The filesystem on /dev/UM4_UG/LV2 is now 786432 (4k) blocks long.

resize2fs done
Extended file system ext4 on UM4_UG/LV2.
Logical volume UM4_UG/LV2 successfully resized.
```

Vérification :

```
[root@UM4 ~]# df -hT /Part1 /Part2
Filesystem      Type  Size  Used Avail Use% Mounted on
/dev/mapper/UM4_UG-LV1 xfs   2.0G  47M  1.9G   3% /Part1
/dev/mapper/UM4_UG-LV2 ext4   2.9G   24K  2.8G   1% /Part2
```

- Donnez le nombre d'inodes et le nombre de blocs de chaque volume logique.

```
[root@UM4 ~]# df -hi /Part1
Filesystem      Inodes IUsed IFree IUse% Mounted on
/dev/mapper/UM4_UG-LV1 1.0M    3 1.0M    1% /Part1
[root@UM4 ~]# df -hi /Part2
Filesystem      Inodes IUsed IFree IUse% Mounted on
/dev/mapper/UM4_UG-LV2 192K   11 192K    1% /Part2
[root@UM4 ~]#
```

```
[root@UM4 ~]# df -hT /Part1 /Part2
Filesystem      Type  Size  Used Avail Use% Mounted on
/dev/mapper/UM4_UG-LV1 xfs   2.0G  47M  1.9G   3% /Part1
/dev/mapper/UM4_UG-LV2 ext4   2.9G   24K  2.8G   1% /Part2
```

PARTIE 2 – SCHÉMA D'ADRESSAGE

Dans cette entreprise, on vous donne l'adresse réseau **172.16.1.0/24** et on vous demande, selon la topologie, de mettre en place un schéma d'adressage répondant aux exigences liées à la taille de chaque réseau.

Créez une **Table de segmentation en sous-réseau** et une **Table des adresses IP** que vous devrez définir en statique pour chaque interface des VMs.

TABLE DE SEGMENTATION

Sous réseau	Masque	Diffusion	Premier hôte	Dernier hôte
TPLlan4	255.255.255.224	172.16.1.31	172.16.1.1	172.16.1.30
TPLlan3	255.255.255.240	172.16.1.47	172.16.1.33	172.16.1.46

TPLlan1	255.255.255.252,	172.16.1.51,	172.16.1.49	172.16.1.50
TPLlan2	255.255.255.252	172.16.1.55,	172.16.1.53	172.16.1.54

TABLE D'ADRESSES IP

VM	Carte Réseau	IP	Réseau virtuel
VM1	NAT	DHCP	NAT
	ens192	172.16.1.49/30	TPLlan1
	ens224	172.16.1.53/30	TPLlan2
VM2	ens160	172.16.1.50/30	TPLlan1
	ens192	172.16.1.33/28	TPLlan3
	ens224	-	-
VM3	ens160	172.16.1.54/30	TPLlan2
VM4	ens160	172.16.1.34/28,	TPLlan3
	ens192	172.16.1.1/27	TPLlan4
	ens224	-	-
VM5	ens160	172.16.1.2/27	TPLlan4

PARTIE 3 – ROUTAGE

- Créez vos **réseaux virtuels** et connectez vos machines selon la topologie donnée en appliquant le schéma d'adressage obtenu dans la partie 2.
- Vérifiez que toutes vos VMs **peuvent communiquer les unes avec les autres** ; sinon, faites en sorte que cela soit possible en **ajoutant des routes statiques** bien identifiées permettant d'aller d'une VM à chaque réseau identifié.

VM1:

```
[root@UM1 ~]# nmcli dev status
```

DEVICE	TYPE	STATE	CONNECTION
ens160	ethernet	connected	ens160
lo	loopback	connected (externally)	lo
ens192	ethernet	disconnected	--
ens224	ethernet	disconnected	--

-Voici les connexions :

Nouvelle connexion LAN1 (TP1Lan1)

```
[root@UM1 ~]# nmcli con add type ethernet ifname ens192 con-name LAN1 ipv4.method manual ipv4.addresses 172.16.1.49/30
Connection 'LAN1' (944c9a40-3c04-4e35-920e-6e7a5410f6fb) successfully added.
```

Nouvelle connexion LAN1 (TP1Lan2)

```
[root@UM1 ~]# nmcli con add type ethernet ifname ens224 con-name LAN2 ipv4.method manual ipv4.addresses 172.16.1.53/30
Connection 'LAN2' (65a01281-8fec-4aad-ad6c-ed069dbb5c03) successfully added.
```

Confirmation des nouvelles connexions :

```
[root@UM1 ~]# nmcli dev status
```

DEVICE	TYPE	STATE	CONNECTION
ens160	ethernet	connected	ens160
ens192	ethernet	connected	LAN1
ens224	ethernet	connected	LAN2
lo	loopback	connected (externally)	lo

Configuration du NAT en DHCP :

```
[root@UM1 ~]# nmcli con add type ethernet ifname ens160 con-name NAT ipv4.method auto
Connection 'NAT' (36e84783-979f-4c30-ab0b-376ac5ca1fc9) successfully added.
```

```
[root@UM1 ~]# nmcli
ens160: connected to ens160
    "VMware VMXNET3"
    ethernet (vmxnet3), 00:0C:29:80:D9:F4, hw, mtu 1500
    ip4 default
    inet4 192.168.85.132/24
    route4 192.168.85.0/24 metric 100
    route4 default via 192.168.85.2 metric 100
    inet6 fe80::20c:29ff:fe80:d9f4/64
    route6 fe80::/64 metric 1024
```

Activer le routage

```
[root@UM1 ~]# sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
-IP Forwarding [root@UM1 ~]# echo "net.ipv4.ip_forward=1" > /etc/sysctl.conf
```

-Zone pare-feu :

```
[root@UM1 ~]# nmcli con mod NAT con.zone external
[root@UM1 ~]# nmcli con mod LAN1 con.zone nm-shared
[root@UM1 ~]# nmcli con mod LAN2 con.zone nm-shared
[root@UM1 ~]# nmcli con d NAT ; nmcli con u NAT
Error: 'NAT' is not an active connection.
Error: no active connection provided.
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
[root@UM1 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/8)
[root@UM1 ~]# nmcli con d LAN2 ; nmcli con u LAN2
Connection 'LAN2' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/9)
```

```
[root@UM1 ~]# nmcli con show NAT
[root@UM1 ~]# nmcli con show NAT | grep zone
connection.zone: external
[root@UM1 ~]# nmcli con show LAN1 | grep zone
connection.zone: nm-shared
[root@UM1 ~]# nmcli con show LAN2 | grep zone
connection.zone: nm-shared
```

Vérifications :

VM2 :

```
[root@UM2 ~]# nmcli dev status
DEVICE  TYPE      STATE                CONNECTION
lo       loopback  connected (externally) lo
ens160   ethernet  disconnected          --
ens192   ethernet  disconnected          --
ens224   ethernet  disconnected          --
```

Voici les connections :

Nouvelle connexion LAN1 (TP1Lan1) :

```
[root@UM2 ~]# nmcli con add type ethernet ifname ens160 con-name LAN1 ipv4.method manual ipv4.addresses 172.16.1.50/30
Connection 'LAN1' (da674293-c84a-4de4-bcda-344ef568c4cf) successfully added.
```

Nouvelle connexion LAN3 (TP1Lan3) :

```
[root@UM2 ~]# nmcli con add type ethernet ifname ens192 con-name LAN3 ipv4.method manual ipv4.addresses 172.16.1.33/30
Connection 'LAN3' (b47c9950-95ed-49e7-a533-24838b3217e2) successfully added.
```

```
[root@UM2 ~]# nmcli dev status
DEVICE  TYPE      STATE                CONNECTION
ens160   ethernet  connected            LAN1
ens192   ethernet  connected            LAN3
lo       loopback  connected (externally) lo
ens224   ethernet  disconnected          --
```

Confirmation des nouvelles connexions :

Activer le routage

```
[root@UM2 ~]# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
```

IP forwarding:

```
[root@UM2 ~]# echo "net.ipv4.ip_forward=1" > /etc/sysctl.conf
```

Pare-feu:

```
[root@UM2 ~]# nmcli con mod LAN1 con.zone nm-shared
[root@UM2 ~]# nmcli con mod LAN3 con.zone nm-shared
[root@UM2 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/15)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/17)
[root@UM2 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/14)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/19)
```

Vérification :

```
[root@UM2 ~]# nmcli con show LAN1 | grep zone
connection.zone: nm-shared
[root@UM2 ~]# nmcli con show LAN3 | grep zone
connection.zone: nm-shared
```

VM3 :

Voici les connections :

```
[root@UM3 ~]# nmcli dev status
DEVICE  TYPE      STATE                CONNECTION
ens160   ethernet  connecting (getting IP configuration) ens160
lo       loopback  connected (externally) lo
[root@UM3 ~]#
```


Nouvelle connexion LAN2 (TP1Lan2) :

```
[root@UM3 ~]# nmcli con add type ethernet ifname ens160 con-name LAN2 ipv4.method manual ipv4.addresses 172.16.1.54/30
Connection 'LAN2' (164b2274-3ad0-47aa-9817-41b66fe49ad0) successfully added.
```

Confirmation des nouvelles connexions :

```
[root@UM3 ~]# nmcli dev status
DEVICE  TYPE      STATE      CONNECTION
ens160  ethernet  connected  LAN2
lo      loopback  connected (externally)  lo
[root@UM3 ~]#
```

VM4 :

Voici les connexions :

```
[root@UM4 ~]# nmcli dev status
DEVICE  TYPE      STATE      CONNECTION
ens160  ethernet  connecting (getting IP configuration)  ens160
ens192  ethernet  connecting (getting IP configuration)  Wired connection 1
lo      loopback  connected (externally)  lo
[root@UM4 ~]#
```

Nouvelle connexion LAN3 :

```
[root@UM4 ~]# nmcli con add type ethernet ifname ens160 con-name LAN3 ipv4.method manual ipv4.addresses 172.16.1.34/28
Connection 'LAN3' (6758069b-ae0e-4a6b-9e23-aff008ac5193) successfully added.
```

Nouvelle connexion LAN4 :



Confirmation des nouvelles connexions :

```
[root@UM4 ~]# nmcli dev status
DEVICE  TYPE      STATE      CONNECTION
ens160  ethernet  connected  LAN3
ens192  ethernet  connected  LAN4
lo      loopback  connected (externally)  lo
[root@UM4 ~]#
```

Activer le routage

```
[root@UM4 ~]# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
[root@UM4 ~]# echo "net.ipv4.ip_forward=1" > /etc/sysctl.conf
```

IP forwarding:

Pare-feu:

```
[root@UM4 ~]# nmcli con mod LAN3 con.zone nm-shared
[root@UM4 ~]# nmcli con mod LAN5 con.zone nm-shared
Error: unknown connection 'LAN5'.
[root@UM4 ~]# nmcli con mod LAN4 con.zone nm-shared
[root@UM4 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/13)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/16)
[root@UM4 ~]# nmcli con d LAN4 ; nmcli con u LAN4
Connection 'LAN4' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/14)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/18)
```

```
[root@VM4 ~]# nmcli con show LAN3 | grep zone
connection.zone: nm-shared
[root@VM4 ~]# nmcli con show LAN4 | grep zone
connection.zone: nm-shared
```

Vérification :

VM5:

Voici les connexions :

```
[root@VM5 ~]# nmcli dev status
DEVICE  TYPE      STATE                                  CONNECTION
ens160  ethernet  connecting (getting IP configuration)  ens160
lo       loopback  connected (externally)                lo
```

Nouvelle connexion LAN

```
[root@VM5 ~]# nmcli con add type ethernet ifname ens160 con-name LAN4 ipv4.method manual ipv4.addresses 172.16.1.2/27
Connection 'LAN4' (97d4b7c6-b584-4237-b5eb-3358201e7e03) successfully added.
```

Confirmation des nouvelles connexions :

```
[root@VM5 ~]# nmcli dev status
DEVICE  TYPE      STATE      CONNECTION
ens160  ethernet  connected  LAN4
lo       loopback  connected (externally)  lo
```

ROUTES STATIQUES

VM1:

Route vers TPLlan3 (172.16.1.32/28) par VM2:

```
[root@VM1 ~]# nmcli con mod LAN1 +ipv4.routes "172.16.1.32/28 172.16.1.50"
[root@VM1 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

#Route vers TPLlan4 (172.16.1.0/27) par VM2:

```
[root@VM1 ~]# nmcli con mod LAN1 +ipv4.routes "172.16.1.0/27 172.16.1.50"
[root@VM1 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
```

Vérification:

```
[root@VM1 ~]# ip route
default via 192.168.85.2 dev ens160 proto dhcp src 192.168.85.132 metric 100
172.16.1.0/27 via 172.16.1.50 dev ens192 proto static metric 103
172.16.1.32/28 via 172.16.1.50 dev ens192 proto static metric 103
172.16.1.48/30 dev ens192 proto kernel scope link src 172.16.1.49 metric 103
172.16.1.52/30 dev ens224 proto kernel scope link src 172.16.1.53 metric 102
192.168.85.0/24 dev ens160 proto kernel scope link src 192.168.85.132 metric 100
```

VM2:

Route vers TPLlan2 (172.16.1.52/30) par VM1

```
[root@VM2 ~]# nmcli con mod LAN1 +ipv4.routes "172.16.1.52/30 172.16.1.49"
[root@VM2 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

Route vers TPLlan4 (172.16.1.0/27) par VM2:

```
[root@VM2 ~]# nmcli con mod LAN3 +ipv4.routes "172.16.1.0/27 172.16.1.34"
[root@VM2 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
```

Vérification:

```
[root@VM2 ~]# ip route
172.16.1.0/27 via 172.16.1.34 dev ens192 proto static metric 103
172.16.1.32/30 dev ens192 proto kernel scope link src 172.16.1.33 metric 103
172.16.1.48/30 dev ens160 proto kernel scope link src 172.16.1.50 metric 102
172.16.1.52/30 via 172.16.1.49 dev ens160 proto static metric 102
```

VM3 :

Route par défaut vers VM1 :

```
[root@VM3 ~]# nmcli con mod LAN2 +ipv4.routes "0.0.0.0/0 172.16.1.53"
[root@VM3 ~]# nmcli con d LAN2 ; nmcli con u LAN2
Connection 'LAN2' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
```

```
[root@VM3 ~]# ip route
default via 172.16.1.53 dev ens160 proto static metric 100
172.16.1.52/30 dev ens160 proto kernel scope link src 172.16.1.54 metric 100
```

Vérification :

VM4 :

Route vers TPLlan1 (172.16.1.48/30) par VM2 :

```
[root@VM4 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

Route vers TPLlan2 (172.16.1.52/30) par VM2

```
[root@VM4 ~]# nmcli con mod LAN3 +ipv4.routes "172.16.1.52/30 172.16.1.33"
[root@VM4 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
```

Vérification :

```
[root@VM4 ~]# ip route
172.16.1.0/27 dev ens192 proto kernel scope link src 172.16.1.1 metric 101
172.16.1.32/28 dev ens160 proto kernel scope link src 172.16.1.34 metric 102
172.16.1.48/30 via 172.16.1.33 dev ens160 proto static metric 102
172.16.1.52/30 via 172.16.1.33 dev ens160 proto static metric 102
```

VM5 :

Route par défaut vers VM4:

```
[root@UM5 ~]# nmcli con mod LAN4 +ipv4.routes "0.0.0.0/0 172.16.1.1"
[root@UM5 ~]# nmcli con d LAN4 ; nmcli con u LAN4
Error: 'LAN4' is not an active connection.
Error: no active connection provided.
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
```

```
[root@UM5 ~]# ip route
default via 172.16.1.1 dev ens160 proto static metric 100
172.16.1.0/27 dev ens160 proto kernel scope link src 172.16.1.2 metric 100
[root@UM5 ~]#
```

Vérification :

PINGS POUR VÉRIFICATION:

```
[root@UM1 ~]# ping 172.16.1.50
PING 172.16.1.50 (172.16.1.50) 56(84) bytes of data.
64 bytes from 172.16.1.50: icmp_seq=1 ttl=64 time=0.705 ms
64 bytes from 172.16.1.50: icmp_seq=2 ttl=64 time=2.36 ms
64 bytes from 172.16.1.50: icmp_seq=3 ttl=64 time=2.64 ms
64 bytes from 172.16.1.50: icmp_seq=4 ttl=64 time=1.98 ms
```

VM1 à VM2 :

```
[root@UM1 ~]# ping 172.16.1.34
PING 172.16.1.34 (172.16.1.34) 56(84) bytes of data.
64 bytes from 172.16.1.34: icmp_seq=1 ttl=63 time=1.03 ms
64 bytes from 172.16.1.34: icmp_seq=2 ttl=63 time=2.32 ms
64 bytes from 172.16.1.34: icmp_seq=3 ttl=63 time=2.41 ms
```

VM1 à VM4 :

```
[root@UM2 ~]# ping 172.16.1.49
PING 172.16.1.49 (172.16.1.49) 56(84) bytes of data.
64 bytes from 172.16.1.49: icmp_seq=1 ttl=64 time=0.644 ms
64 bytes from 172.16.1.49: icmp_seq=2 ttl=64 time=1.87 ms
64 bytes from 172.16.1.49: icmp_seq=3 ttl=64 time=1.54 ms
64 bytes from 172.16.1.49: icmp_seq=4 ttl=64 time=1.73 ms
```

VM2 à VM1 :

```
[root@UM2 ~]# ping 172.16.1.2
PING 172.16.1.2 (172.16.1.2) 56(84) bytes of data.
64 bytes from 172.16.1.2: icmp_seq=1 ttl=63 time=1.12 ms
64 bytes from 172.16.1.2: icmp_seq=2 ttl=63 time=1.27 ms
64 bytes from 172.16.1.2: icmp_seq=3 ttl=63 time=3.25 ms
```

VM2 à VM5 :

```
[root@UM5 ~]# ping 172.16.1.1
PING 172.16.1.1 (172.16.1.1) 56(84) bytes of data.
64 bytes from 172.16.1.1: icmp_seq=1 ttl=64 time=0.521 ms
64 bytes from 172.16.1.1: icmp_seq=2 ttl=64 time=1.25 ms
64 bytes from 172.16.1.1: icmp_seq=3 ttl=64 time=1.98 ms
```

VM5 à VM4 :

```
[root@VM5 ~]# ping 172.16.1.49
PING 172.16.1.49 (172.16.1.49) 56(84) bytes of data.
64 bytes from 172.16.1.49: icmp_seq=1 ttl=62 time=1.56 ms
64 bytes from 172.16.1.49: icmp_seq=2 ttl=62 time=5.04 ms
64 bytes from 172.16.1.49: icmp_seq=3 ttl=62 time=2.54 ms
64 bytes from 172.16.1.49: icmp_seq=4 ttl=62 time=4.52 ms
```

VM5 À VM1 :

```
[root@VM5 ~]# ping 172.16.1.54
PING 172.16.1.54 (172.16.1.54) 56(84) bytes of data.
64 bytes from 172.16.1.54: icmp_seq=1 ttl=61 time=1.66 ms
64 bytes from 172.16.1.54: icmp_seq=2 ttl=61 time=5.89 ms
64 bytes from 172.16.1.54: icmp_seq=3 ttl=61 time=2.79 ms
```

VM5 À VM3 :

```
[root@VM4 ~]# ping 172.16.1.33
PING 172.16.1.33 (172.16.1.33) 56(84) bytes of data.
64 bytes from 172.16.1.33: icmp_seq=1 ttl=64 time=0.758 ms
64 bytes from 172.16.1.33: icmp_seq=2 ttl=64 time=2.77 ms
64 bytes from 172.16.1.33: icmp_seq=3 ttl=64 time=1.99 ms
```

VM4 À VM2 :

```
[root@VM4 ~]# ping 172.16.1.54
PING 172.16.1.54 (172.16.1.54) 56(84) bytes of data.
64 bytes from 172.16.1.54: icmp_seq=1 ttl=62 time=2.29 ms
64 bytes from 172.16.1.54: icmp_seq=2 ttl=62 time=5.87 ms
64 bytes from 172.16.1.54: icmp_seq=3 ttl=62 time=0.918 ms
```

VM4 À VM3 :

```
[root@VM3 ~]# ping 172.16.1.2
PING 172.16.1.2 (172.16.1.2) 56(84) bytes of data.
64 bytes from 172.16.1.2: icmp_seq=1 ttl=61 time=2.01 ms
64 bytes from 172.16.1.2: icmp_seq=2 ttl=61 time=5.55 ms
64 bytes from 172.16.1.2: icmp_seq=3 ttl=61 time=6.13 ms
64 bytes from 172.16.1.2: icmp_seq=4 ttl=61 time=3.38 ms
```

VM3 À VM5 :

PARTIE 4 – DÉFI (Obligatoire)

Faites-en sorte que vos VMs puissent aller sur Internet.

Astuce : Vous pouvez tester la connexion Internet avec la commande **dnf install ftp** si les requêtes ping sont bloquées.

Test de connexion sur la VM1:

```
[root@VM1 ~]# dnf install ftp
Last metadata expiration check: 0:35:11 ago on Thu Oct 30 15:07:27 2025.
Dependencies resolved.
=====
Package                                Architecture      Version
=====
Installing:
ftp                                     x86_64            0.17-89.e19

Transaction Summary
=====
Install 1 Package

Total download size: 61 k
Installed size: 112 k
Is this ok [y/N]: y
Downloading Packages:
ftp-0.17-89.e19.x86_64.rpm
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 
  Installing     : ftp-0.17-89.e19.x86_64
  Running scriptlet: ftp-0.17-89.e19.x86_64
  Verifying      : ftp-0.17-89.e19.x86_64

Installed:
ftp-0.17-89.e19.x86_64

Complete!
```

VM2 :

Ajouter la route par défaut vers VM1:

```
[root@VM2 ~]# nmcli con mod LAN1 +ipv4.routes "0.0.0.0/0 172.16.1.49"
[root@VM2 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

Vérification :

```
[root@VM2 ~]# ip route
default via 172.16.1.49 dev ens160 proto static metric 102
172.16.1.0/27 via 172.16.1.34 dev ens192 proto static metric 101
172.16.1.32/30 dev ens192 proto kernel scope link src 172.16.1.33 metric 101
172.16.1.48/30 dev ens160 proto kernel scope link src 172.16.1.50 metric 102
172.16.1.52/30 via 172.16.1.49 dev ens160 proto static metric 102
```

VM4 : Ajouter la route par défaut vers VM2:

```
[root@VM4 ~]# nmcli con mod LAN3 +ipv4.routes "0.0.0.0/0 172.16.1.33"
[root@VM4 ~]# nmcli con d LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
```

Vérification :

```
[root@VM1 ~]# ip route
default via 172.16.1.33 dev ens160 proto static metric 102
172.16.1.0/27 dev ens192 proto kernel scope link src 172.16.1.1 metric 101
172.16.1.32/28 dev ens160 proto kernel scope link src 172.16.1.34 metric 102
172.16.1.48/30 via 172.16.1.33 dev ens160 proto static metric 102
172.16.1.52/30 via 172.16.1.33 dev ens160 proto static metric 102
```

CONFIGURATION DU DNS SUR LES VM2 À VM5 POUR FAIRE LE TEST DE CONNEXION

VM2

DNS :

```
[root@VM2 ~]# nmcli con mod LAN1 ipv4.dns "192.168.85.2"
[root@VM2 ~]# nmcli con d LAN1 ; nmcli con u LAN1
Connection 'LAN1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/9)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/11)
```

Test de connexion sur la VM2:

```
[root@VM2 ~]# dnf install ftp -y
AlmaLinux 9 - AppStream                               34 kB/s | 4.2 kB      00:00
^CAlmaLinux 9 - AppStream                               1 320 kB/s | 2.1 MB    01:00 E
AlmaLinux 9 - AppStream                               451 kB/s | 2.1 MB     00:04
Error: Failed to download metadata for repo 'appstream': Librepo was interrupted by a signal
[root@VM2 ~]# dnf install ftp -y
AlmaLinux 9 - AppStream                               32 kB/s | 4.2 kB      00:00
AlmaLinux 9 - AppStream                               14 MB/s | 21 MB       00:01
AlmaLinux 9 - BaseOS                                  30 kB/s | 3.8 kB      00:00
AlmaLinux 9 - BaseOS                                  12 MB/s | 33 MB       00:02
AlmaLinux 9 - Extras                                  33 kB/s | 3.3 kB      00:00
AlmaLinux 9 - Extras                                  156 kB/s | 20 kB      00:00
Dependencies resolved.
=====
Package           Architecture      Version           Repository        Size
=====
Installing:
ftp                x86_64            0.17-89.e19      appstream         61 k
Transaction Summary
=====
Install 1 Package

Total download size: 61 k
Installed size: 112 k
Downloading Packages:
ftp-0.17-89.e19.x86_64.rpm                               303 kB/s | 61 kB      00:00
-----
Total                                                    213 kB/s | 61 kB      00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :
  Installing                : ftp-0.17-89.e19.x86_64                1/1
  Running scriptlet         : ftp-0.17-89.e19.x86_64                1/1
  Verifying                 : ftp-0.17-89.e19.x86_64                1/1
Installed:
ftp-0.17-89.e19.x86_64
Complete!
```

VM3

DNS:

```
[root@VM3 ~]# nmcli con mod LAN2 ipv4.dns "192.168.85.2"
[root@VM3 ~]# nmcli con d LAN2 ; nmcli con u LAN2
Connection 'LAN2' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
```

Test de connexion sur la VM3:

```
[root@VM3 ~]# dnf install ftp
Last metadata expiration check: 20:37:27 ago on Thu Oct 30 15:39:54 2025.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
ftp                            x86_64            0.17-89.el9       appstream         61 k

Transaction Summary
=====
Install 1 Package

Total download size: 61 k
Installed size: 112 k
Is this ok [y/N]: y
Downloading Packages:
ftp-0.17-89.el9.x86_64.rpm                                           784 kB/s | 61 kB    00:00
-----
Total                                                                    275 kB/s | 61 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                             1/1
  Installing     : ftp-0.17-89.el9.x86_64                      1/1
  Running scriptlet: ftp-0.17-89.el9.x86_64                      1/1
  Verifying      : ftp-0.17-89.el9.x86_64                      1/1

Installed:
ftp-0.17-89.el9.x86_64

Complete!
```

VM4

DNS:

```
[root@VM4 ~]# nmcli con mod LAN3 ipv4.dns "192.168.85.2"
[root@VM4 ~]# nmcli con down LAN3 ; nmcli con u LAN3
Connection 'LAN3' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

Test de connexion sur la VM4:

```
[root@VM4 ~]# sudo dnf install ftp
AlmaLinux 9 - AppStream                28 kB/s | 4.2 kB    00:00
AlmaLinux 9 - AppStream                4.6 MB/s | 21 MB    00:04
AlmaLinux 9 - BaseOS                   38 kB/s | 3.8 kB    00:00
AlmaLinux 9 - BaseOS                   7.5 MB/s | 33 MB    00:04
AlmaLinux 9 - Extras                   38 kB/s | 3.3 kB    00:00
AlmaLinux 9 - Extras                   215 kB/s | 20 kB    00:00
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
ftp                            x86_64            0.17-89.el9       appstream         61 k

Transaction Summary
=====
Install 1 Package

Total download size: 61 k
Installed size: 112 k
Is this ok [y/N]: y
Downloading Packages:
ftp-0.17-89.el9.x86_64.rpm                                           3.6 MB/s | 61 kB    00:00
-----
Total                                                                    568 kB/s | 61 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                             1/1
  Installing     : ftp-0.17-89.el9.x86_64                      1/1
  Running scriptlet: ftp-0.17-89.el9.x86_64                      1/1
  Verifying      : ftp-0.17-89.el9.x86_64                      1/1

Installed:
ftp-0.17-89.el9.x86_64

Complete!
```

VM5

DNS:

```

Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/1)
[root@VM5 ~]# nmcli con mod LAN4 ipv4.dns "192.168.85.2"
[root@VM5 ~]# nmcli con d LAN4 ; nmcli con u LAN4
Connection 'LAN4' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)

```

Test de connexion sur la VM5:

```

[root@VM5 ~]# dnf install ftp
AlmaLinux 9 - AppStream                               35 kB/s | 4.2 kB   00:00
AlmaLinux 9 - AppStream                               12 MB/s | 21 MB   00:01
AlmaLinux 9 - BaseOS                                   43 kB/s | 3.8 kB   00:00
AlmaLinux 9 - BaseOS                                   12 MB/s | 33 MB   00:02
AlmaLinux 9 - Extras                                   37 kB/s | 3.3 kB   00:00
AlmaLinux 9 - Extras                                   197 kB/s | 20 kB   00:00
Dependencies resolved.
=====
Package                        Architecture      Version           Repository         Size
=====
Installing:
ftp                            x86_64            0.17-89.el9       appstream           61 k
=====
Transaction Summary
=====
Install 1 Package

Total download size: 61 k
Installed size: 112 k
Is this ok [y/N]: y
Downloading Packages:
ftp-0.17-89.el9.x86_64.rpm                                           3.5 MB/s | 61 kB   00:00
-----
Total                                                                577 kB/s | 61 kB   00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : ftp-0.17-89.el9.x86_64
  Running scriptlet: ftp-0.17-89.el9.x86_64
  Verifying      : ftp-0.17-89.el9.x86_64
                                           1/1
                                           1/1
                                           1/1
                                           1/1

Installed:
ftp-0.17-89.el9.x86_64

Completed

```